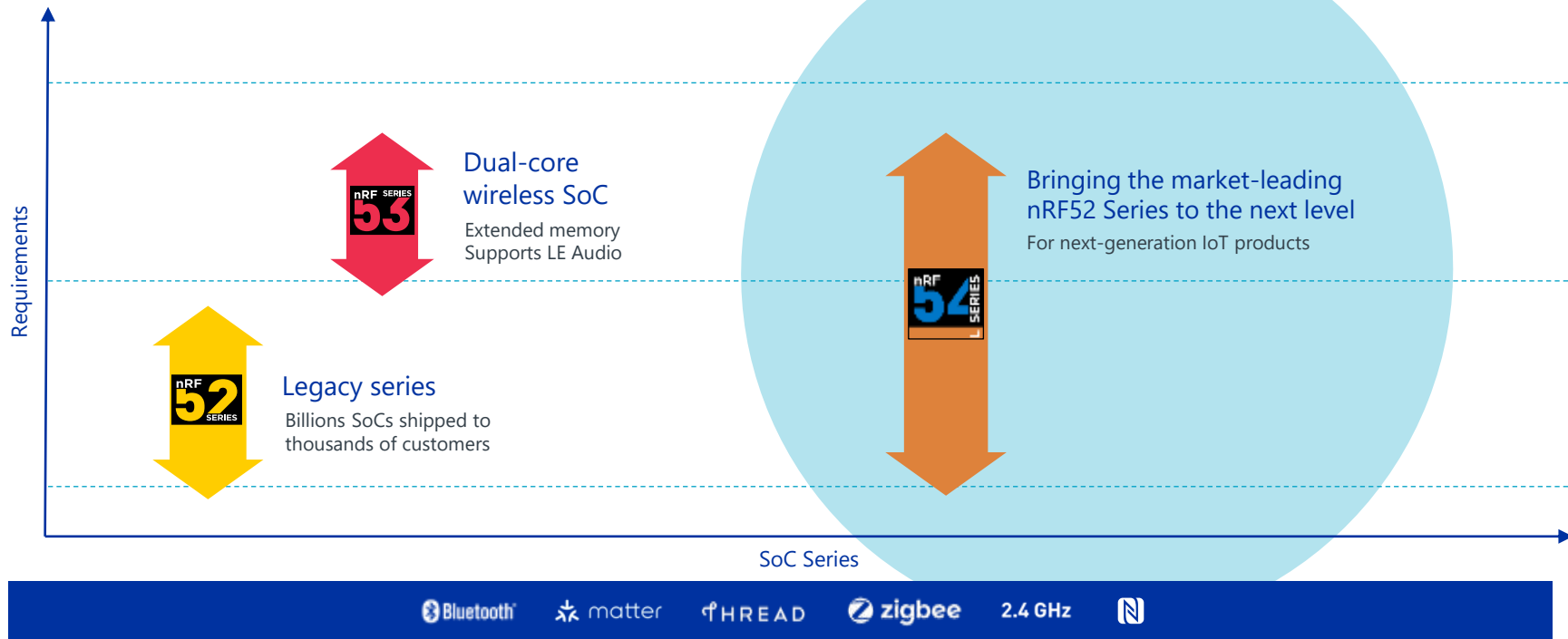


nRF54L Series Architecture

Product portfolio

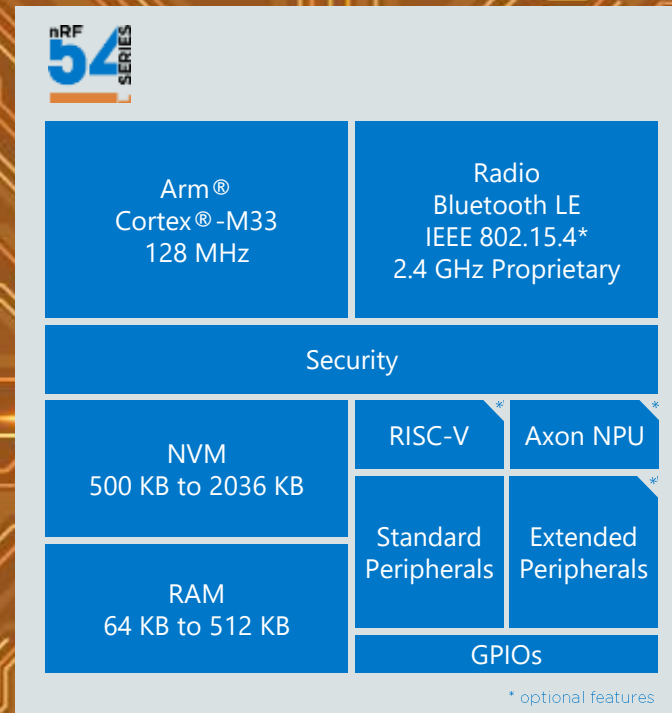
nRF54 Series – Next level wireless SoCs



The nRF54L Series

Architecture and overview

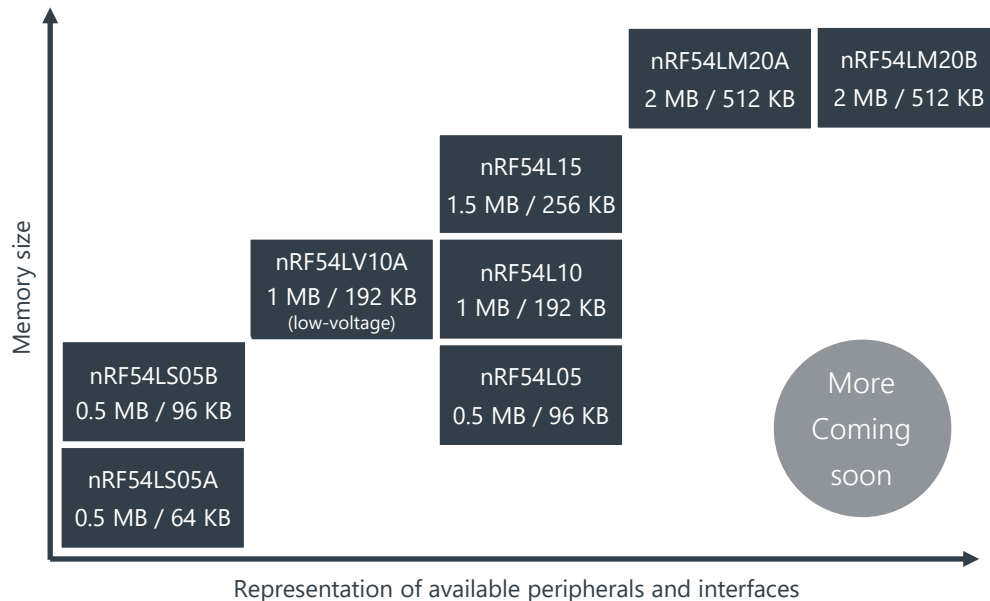
- New technologies enabling the next-level performance
 - 2x processing performance and feature-rich MCU
 - 4th generation radio, with Bluetooth Channel Sounding
 - Ultra-low-power design and 22nm low-leakage process node
- Extended battery life
 - 3x processing efficiency
 - Significantly reduced TX and RX system current
 - Up to 50% lower total power consumption
- Built-in security
 - Provides robust protection against threats
 - Enables compliance with regulations



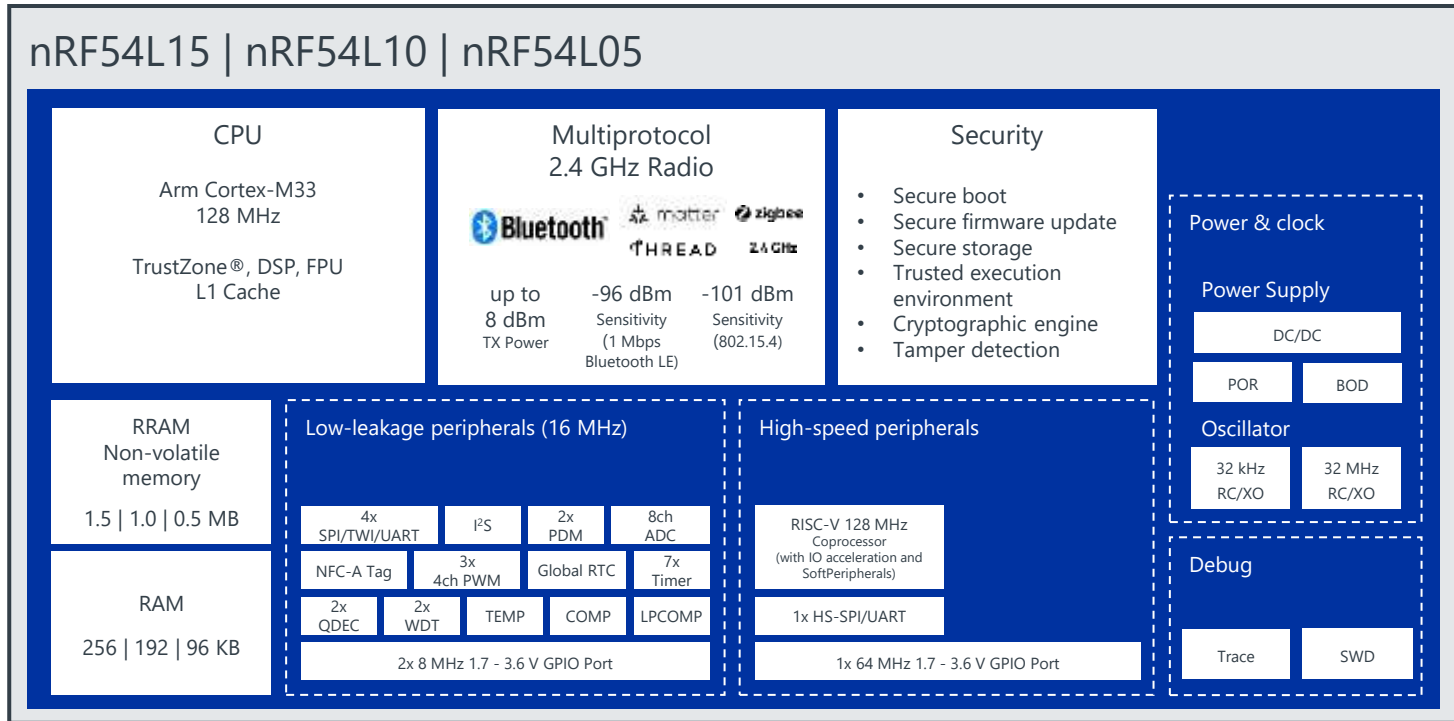
The nRF54L Series

Broad selection of features

- Flexible choice of memory sizes
 - 0.5 MB up to 2 MB NVM
 - 64 KB to 512 KB RAM
- Range of features, peripherals, and interfaces available in various SoC options
 - Extended set of peripherals including High-speed USB, ADC, TDM / I2S, High-speed SPI / UART
 - 128 MHz RISC-V coprocessor and SoftPeripherals
 - 128 MHz Axon NPU AI accelerator
 - Support for silver-oxide batteries and hibernation mode



Block Diagram





nRF54L Deep-Dive

GPIO Ports and Pin Planning

nRF54L Series GPIO Ports

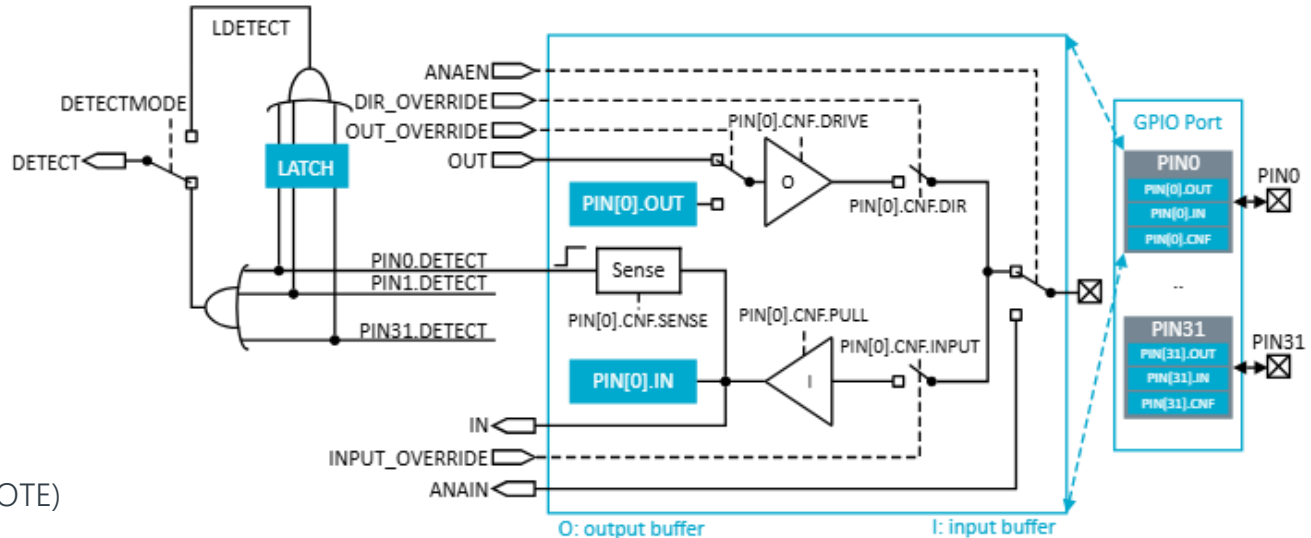
Same as
in nRF52
Series!!

Basic GPIO Port support:

- Pin Direction
- Pull-up and pull-down
- Drive strength

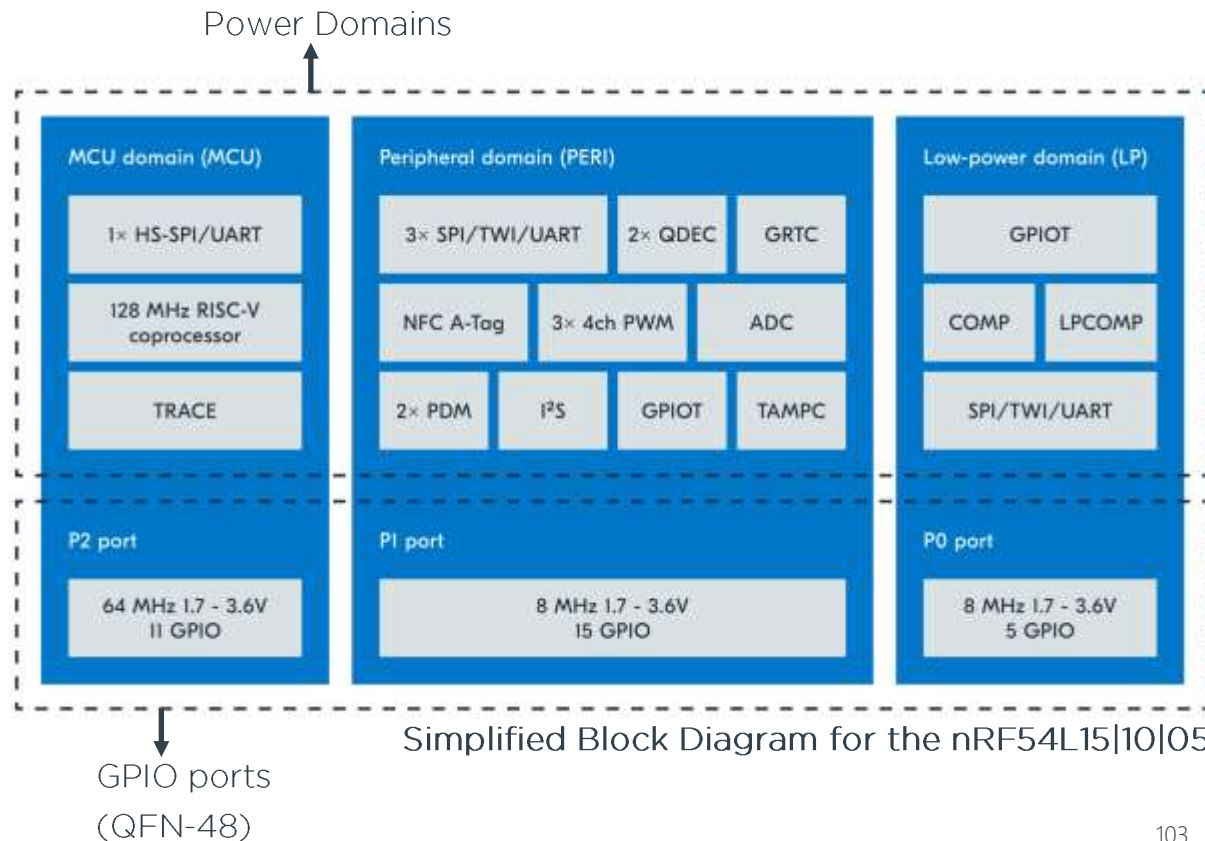
Advanced features:

- Analog pin
- Wake-up source
- Tasks & Events (through GPIOTE)
- Pin sense mechanism



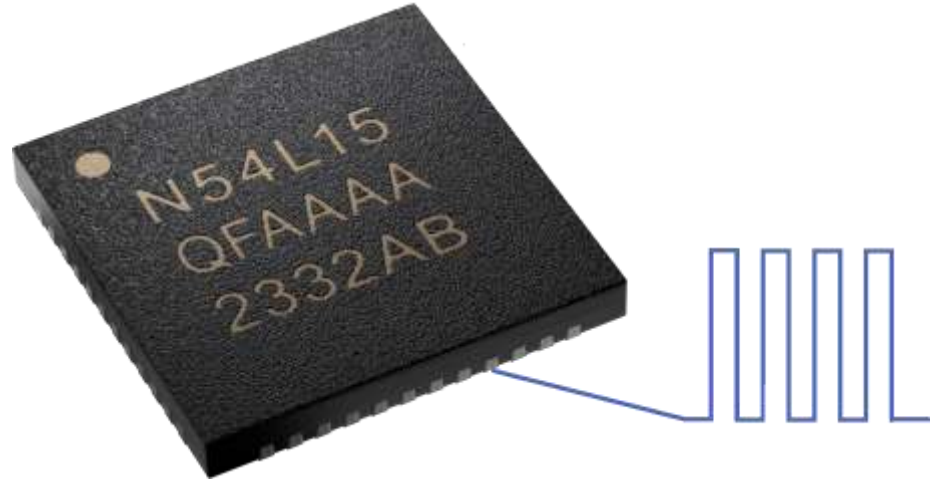
nRF54L Series GPIO Ports

- Unified I/O voltage: 1.7 V to 3.6 V
- Splitting into high-speed (64 MHz) and low-speed ports (8 MHz)
- MCU domain GPIO Port:
 - maximum speed for signals (64 MHz)
 - standard, high, and extra high drive
 - no system wake-up
- PERI and LP domain GPIO Port:
 - power optimized (8 MHz)
 - standard, and high drive
 - wake-up system (System ON or OFF)



... but there is more

- Some peripherals output high-speed clock signals
- Dedicated clock pins have been optimized to ensure correct timing relationship between clock and data signals for these peripherals



Clock Pins

[Example: From the nRF54L15 datasheet](#)

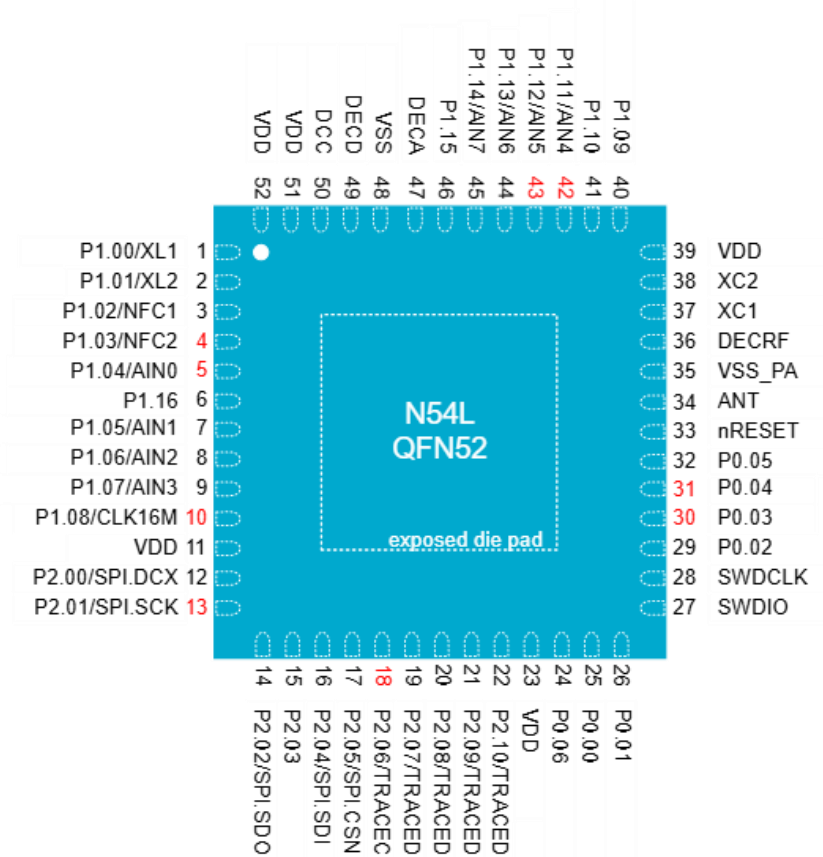
List of peripheral signals and clock pin requirement

Peripheral	Signal	Clock pin required
SPIM/SPIS	SDO	
	SDI	
	SCK	Yes
	CSN	
	DCX	
TWIM/TWIS	SCL	Yes
	SDA	
PDM	DIN	
	CLK	Yes
I2S	MCK	Yes
	LRCK	
	SCK	Yes
	SDIN	
	SDOUT	
TRACE	TRACEDATA[]	
	TRACECLK	Yes (dedicated pin)
GRTC	CLKOUT32K	Yes (dedicated pin)
	PWMOUT	Yes (dedicated pin)
	CLKOUTFAST	Yes (dedicated pin)

Clock Pins

[Example: From the nRF54L15 datasheet](#)

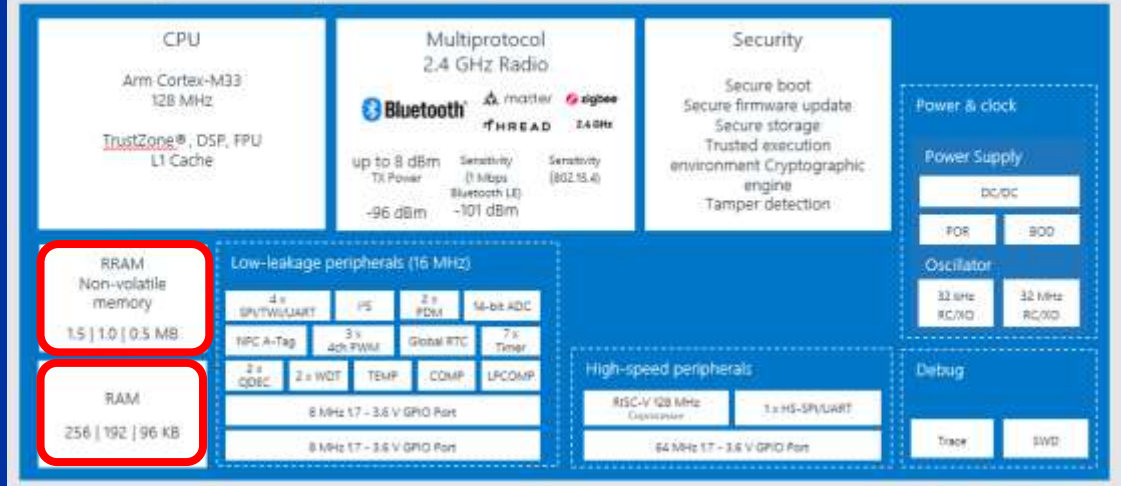
Pins that can be used as clock signals are shown in red in the figure. For more information about clock pins, see Clock pins.



nRF54L Pin Planner (<https://pinplanner.app>)

The screenshot displays the nRF54L Pin Planner web application. The interface is divided into several sections:

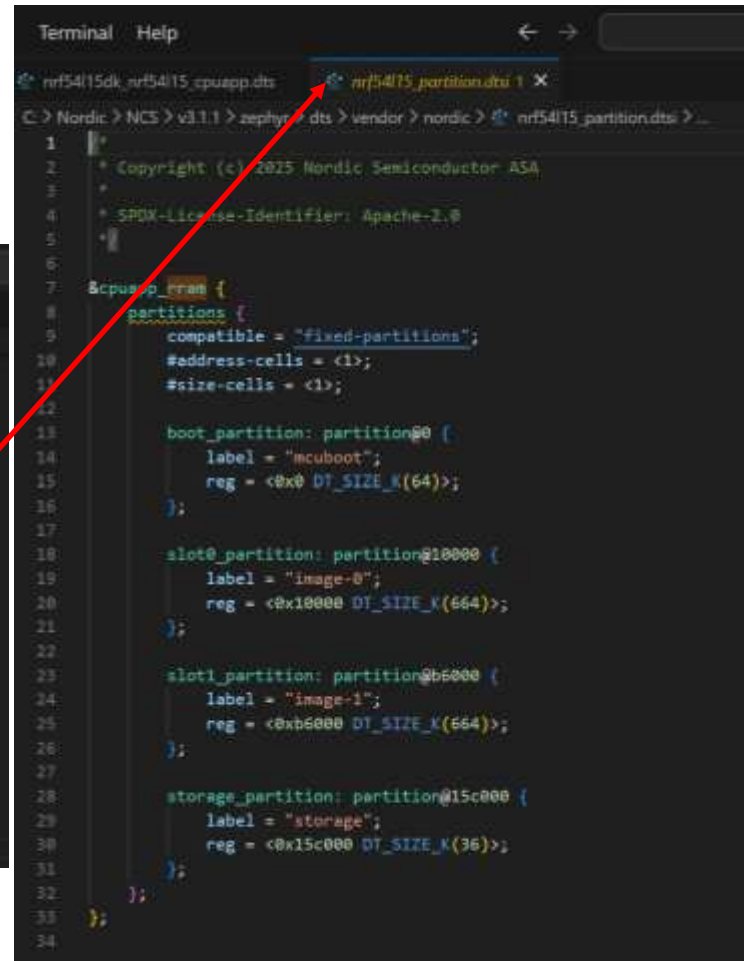
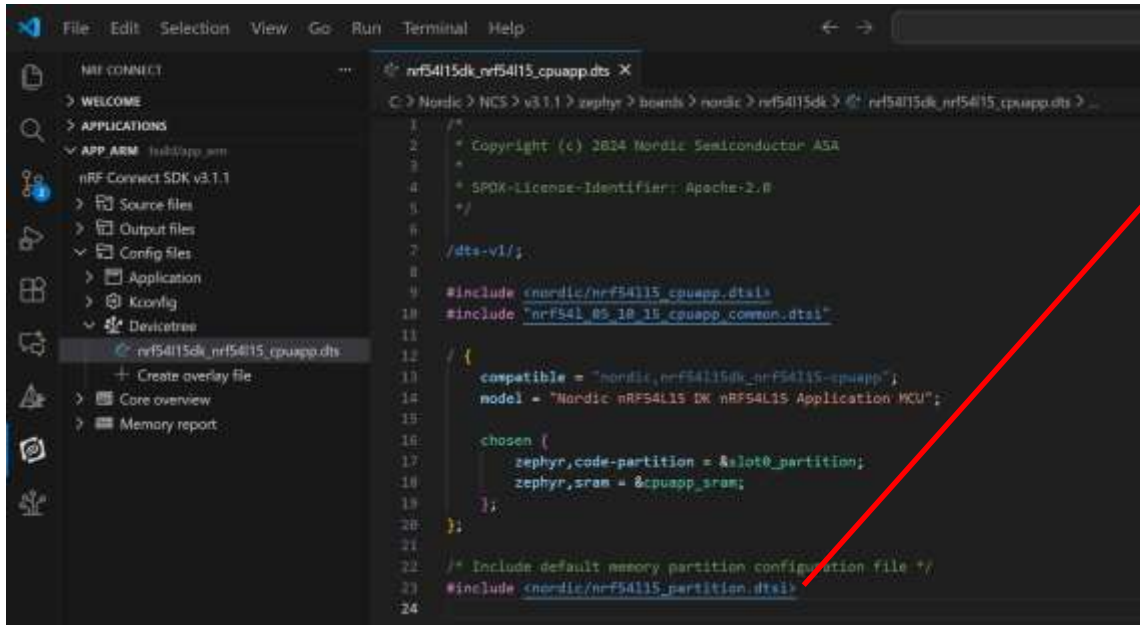
- Configuration:** Includes dropdown menus for the MCU (nRF54LM2DA) and Package (FCCSP-98 (1.7x1.8)).
- Peripherals:** A search bar and a list of peripherals with checkboxes. The list includes:
 - High Frequency Crystal (1x1.6mm) (checked)
 - Low Frequency Crystal Oscillator (Add)
 - NFC Active Tag (Uses P1.01, P1.02)
 - Quad Serial Peripheral Interface (Uses P2.07, P2.08, P2.09, P2.34, P2.05, P2.06)
 - High-speed USB 2.0 device (Uses D+, D-)
 - RTC (RTC)
 - RADIO (RADIO)
 - SAADC (SAADC)
- FCCSP-98 Pin Layout:** A grid of 98 pins. A legend below the grid indicates pin status: Available (grey), Used (orange), Clock Pin (blue), and Power/Debug (green).
- Selected:** A section showing the selected peripheral, HX00, with a note "Internal clock, 16.00 MHz". It includes buttons for "Clear All" and "Generate Device Tree (DIT.DT)".



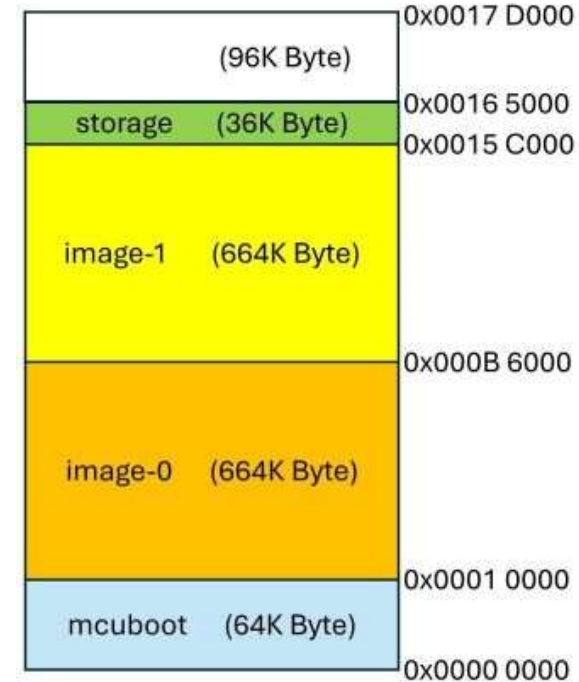
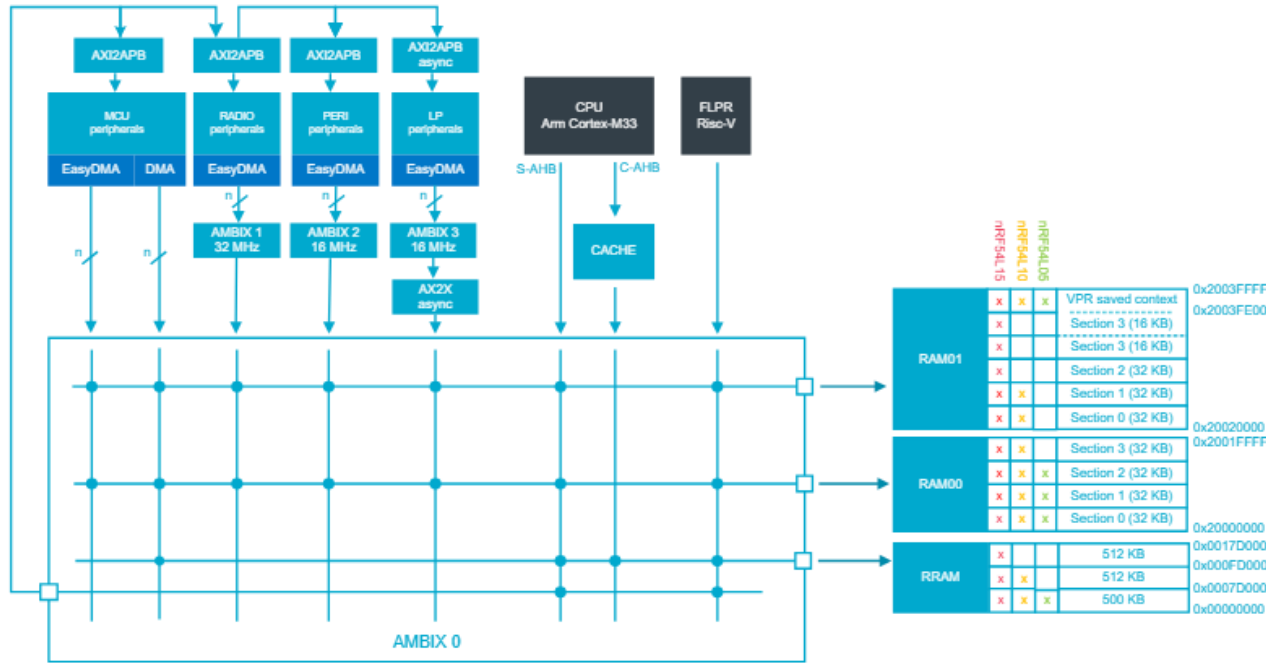
nRF54L Deep-Dive

Memory Map

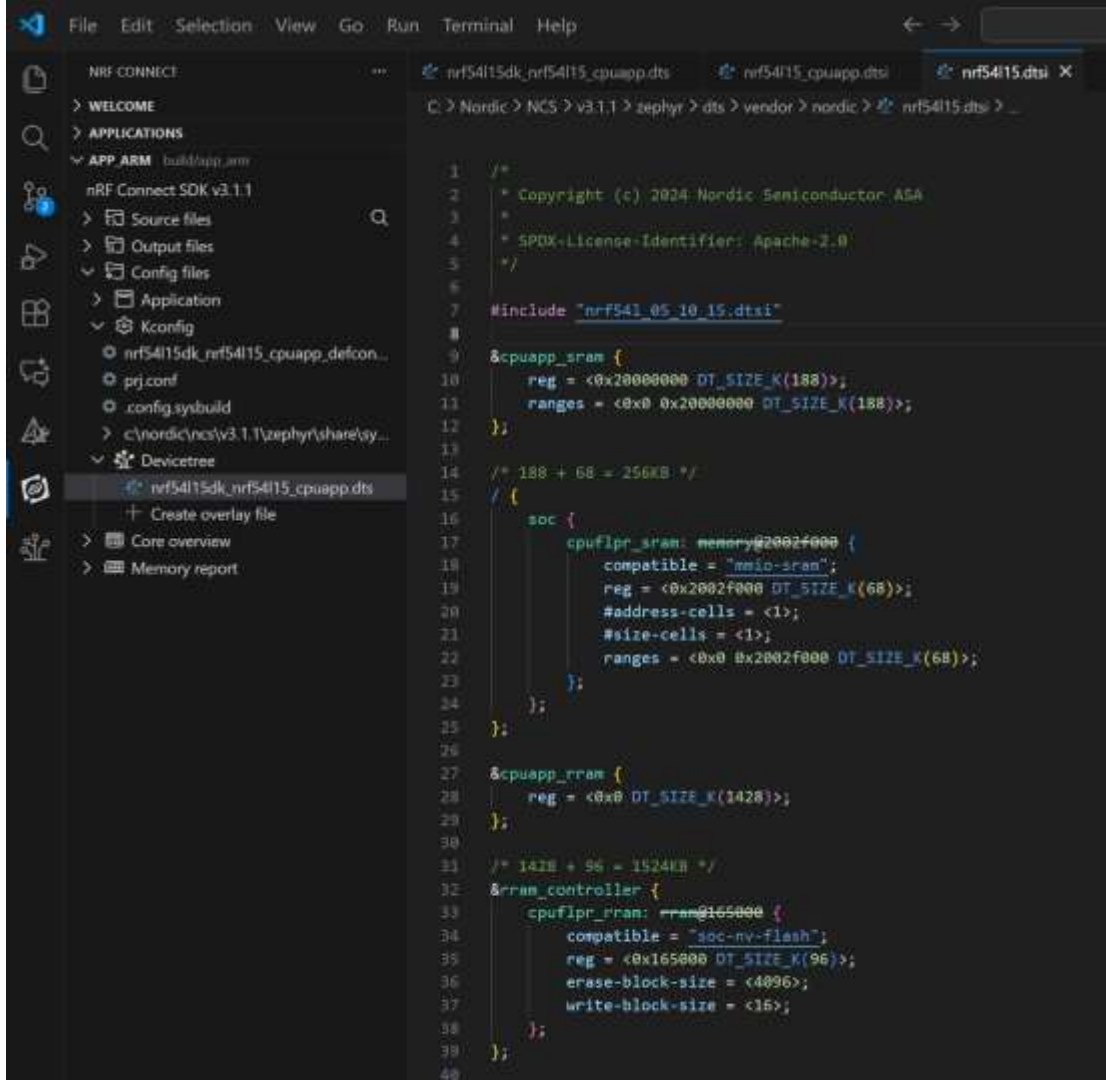
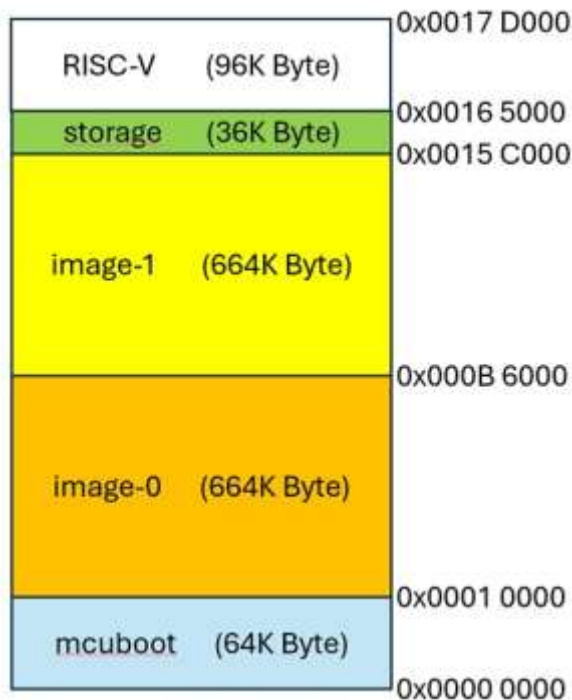
nRF54L15DK Board Definition



nRF54L15 Memory Map vs. DTS file



RISC-V?



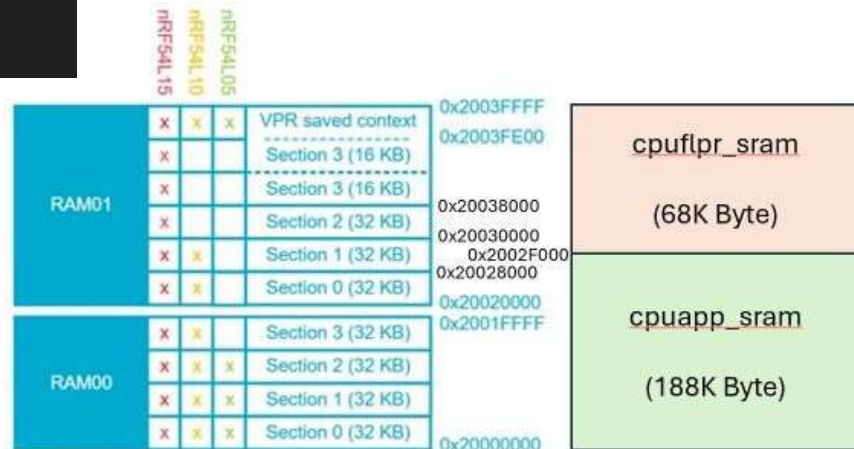
SRAM

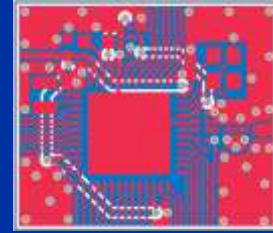
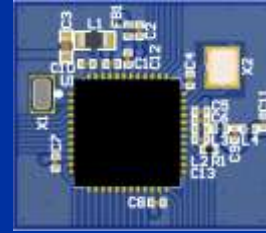
```

9  &cpuapp_sram {
10     reg = <0x20004000 DT_SIZE_K(188)>;
11     ranges = <0x0 0x20000000 DT_SIZE_K(188)>;
12 };
13
14 /* 188 + 68 = 256KB */
15 {
16     soc {
17         cpufldr_sram: memory@2002f000 {
18             compatible = "nisa-sram";
19             reg = <0x2002f000 DT_SIZE_K(68)>;
20             #address-cells = <1>;
21             #size-cells = <1>;
22             ranges = <0x0 0x2002f000 DT_SIZE_K(68)>;
23         };
24     };
25 };
26

```

Each RAM region has separate power control for System ON and System OFF mode. This preserves RAM contents in sleep modes or powers off RAM to save power. The sections are illustrated in [Memory layout](#), and the register interface is described in [MEMCONF — Memory configuration](#).





nRF54L Deep-Dive

Hardware Layout and Design

nRF54L15 Reference Layout

<https://www.nordicsemi.com/Products/nRF54L15/Reference-layout?lang=en#infotabs>

QFN52 package

Applicable for: nRF54L15,
nRF54L10, and nRF54L05

Selected version:
1.0

→ nRF54L15-QFN52 Reference Layout 1.0.zip

Changelog:

1.0

→ The first public version

QFN48 package

Applicable for: nRF54L15,
nRF54L10, and nRF54L05

Selected version:
0.8

→ nRF54L15-QFN48 Reference Layout 0.8.zip

Altium Designer files

Bill of Materials

Schematic and Layout pdf files

Changelog:

0.8

→ The first public version

CSP47 package

Applicable for: nRF54L15

Selected version:
0.7

→ nRF54L15-CSP47 Reference Layout 0.7.zip

Changelog:

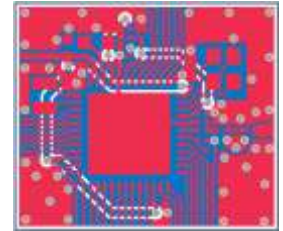
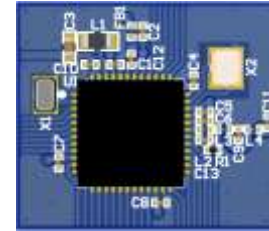
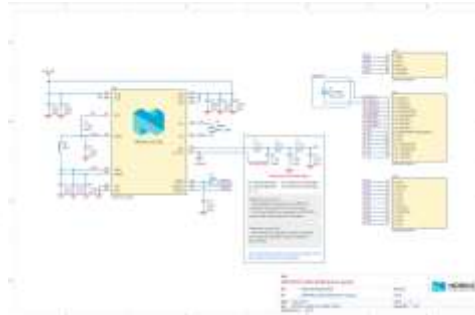
0.7

→ First version of the CSP47 layout
→ Detailed notes: Note regarding the CSP47 layout (0.7) is in the release
→ Please refer to the notes for details (0.7) and (0.8)

0.6

0.5

0.4



Datasheet

Reference layout

Download the reference layout files for the nRF54L15-QFN52 package. The files are available in the following formats:

- Altium Designer files
- Bill of Materials
- Schematic and Layout pdf files

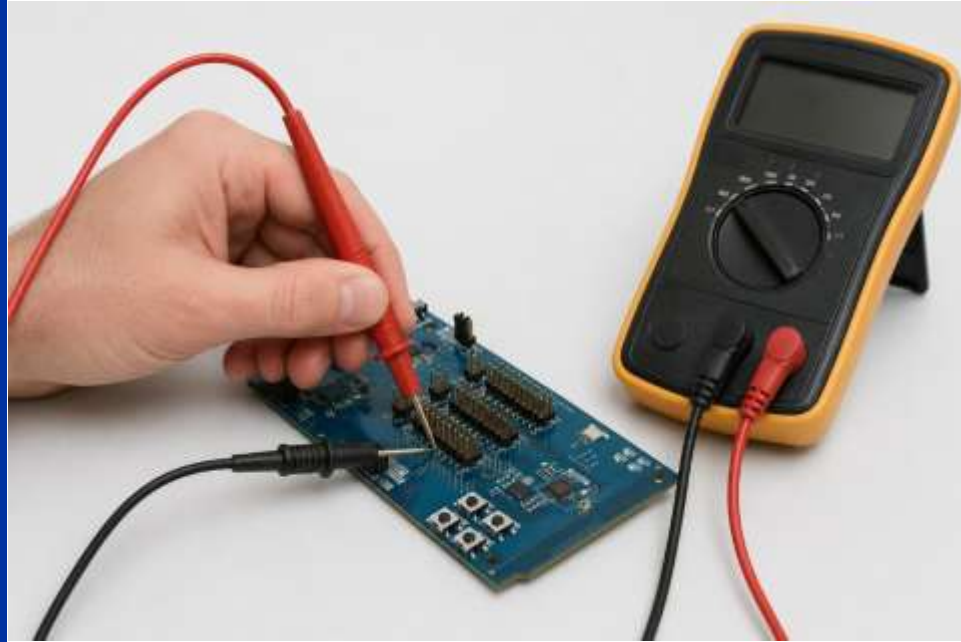
Circuit configuration 1 for QFN52 (QFN52)

Download the circuit configuration 1 for QFN52 (QFN52) files. The files are available in the following formats:

- Altium Designer files
- Bill of Materials
- Schematic and Layout pdf files

Hands-on

- Using SPI





Step 1 – Zephyr SPI Driver

- 1) Copy sample *SPI Bitbang Driver Sample*
- 2) Build the project.
- 3) Check which Zephyr SPI drivers are included?

Step 2 – NRFX SPI Driver

1. Let's take a look into the `nrfx_spim_non-blocking` sample

Please open the project in the SDK folder:

```
C:/Nordic/NCS/v3.4.0/modules/hal/nordic/nrfx/samples/src/nrfx_spim/non_blocking
```